TRAINER KEY. DO NOT DISTRIBUTE. Day 1, Session 2: Good news. Enough to act?

Source: H2-A. Key figures: participants 1,432 to 763 (-669), 4,794 businesses; comparison 2,258 to 2,401 (+143), 5,206 businesses; after-programme gap -1,638; before-and-after interval -684 to -654, p < 0.001. Rule: reduction of at least 1,000 AED per business per year, caused by the programme. Neither comparison alone shows what GreenWaste caused. Decision cards have no single right placement: judge the reason, and ask which action is meant (acting on a descriptive finding can mean "investigate", not "expand").

Underlined = claim the source supports. ~~Struck out~~ = claim the source does not support. Green boxes = model answers (accept equivalent wording).

Page 1 | H2-A: The city data tables and charts

GreenWaste offers equipment subsidies, installation help and staff training. All costs are annual waste management cost, in AED per business, measured before GreenWaste and 12 months after. GreenWaste is a fictional training case.

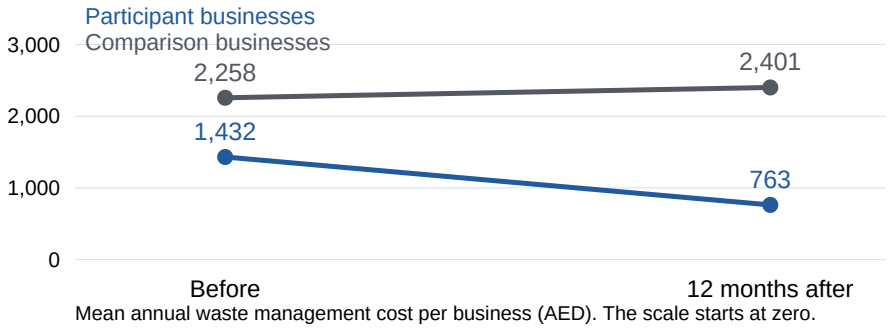
H2-A1: Four selected businesses: efficiency score, whether the business took part, and annual waste management cost before and 12 months after (AED)

Business	Efficiency score	Participation	Before (AED)	After (AED)
B00401	43.6	Took part	2,338	1,629
B00402	88.3	Did not take part	2,425	2,897
B00403	42.4	Took part	1,460	482
B00404	86.6	Did not take part	1,756	2,007

These four businesses were selected from the GreenWaste data for reading practice. They are not representative of all city businesses. Businesses with a score of 58 or below took part in GreenWaste, and businesses with a score above 58 did not. The pilot lottery uses a different set of businesses.

H2-A2: Mean annual waste management cost before and 12 months after GreenWaste, for participant businesses and comparison businesses (AED)

Group	Businesses	Before (AED)	After (AED)	Change (AED)
Participant businesses (took part)	4,794	1,432	763	-669
Comparison businesses (did not take part)	5,206	2,258	2,401	143



Mean annual waste management cost per business (AED). The scale starts at zero.

Each mean uses all the businesses in that group. Figures are rounded. A lottery did not decide which businesses took part, and the two groups had different costs before GreenWaste.

Before and after: for the same participant businesses, the mean cost after minus the mean cost before is **-669 AED**.

With and without: after the programme, the participant businesses' mean cost minus the comparison businesses' mean cost is **-1,638 AED**.

Neither comparison alone shows how much of the fall GreenWaste caused. Our rule is a reduction in annual waste management cost of at least 1,000 AED per business per year, caused by the programme.

Source key

Source: GreenWaste city data (10,000 businesses). Decision cards 2 and 3 use a before-and-after regression for the same 4,794 participant businesses, measured twice. It shows a change of **-669 AED**, with a 95% interval of **-684 to**

-654 AED and **p < 0.001**. The interval describes the mean change. It does not describe individual businesses, and it does not show the effect of the programme.

Page 2 | Meet the records and the distribution

Opening judgement

The mean annual waste management cost of participant businesses fell by 669 AED. Does this tell us the effect of the programme? Write one reason. We will come back to it at the end of the session.

> Example: No. The same businesses may have been affected by something else over the year (prices, weather, other programmes), so a before-and-after fall does not show what would have happened without GreenWaste.

Q1 | Describe one business and the group it belongs to

Use H2-A1 on page 1.

a. Write one sentence about business B00402. Say whether it took part, what its cost was before and after, and over what period.

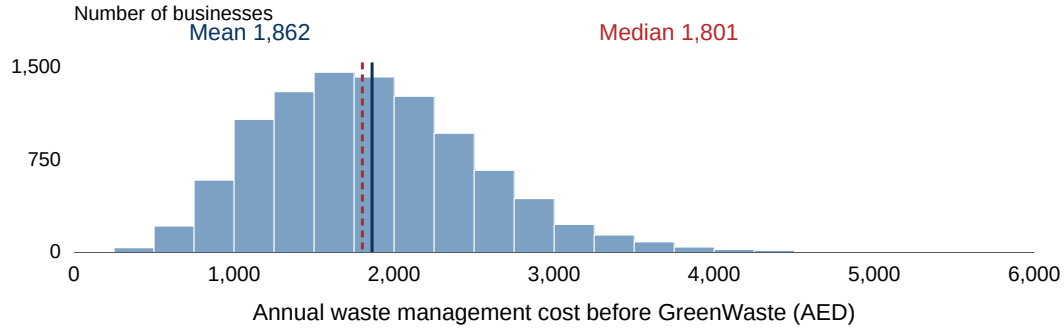
> Example: B00402 did not take part (score 88.3); its annual waste management cost rose from 2,425 AED before to 2,897 AED 12 months after.

b. Which group of businesses does the participant average describe, and how many businesses are in it?

> The 4,794 participant businesses (city businesses with a score of 58 or below), not all 10,000. Not the pilot.

H2-A: Costs before GreenWaste

Histogram of annual waste management cost before GreenWaste for all 10,000 city businesses (AED, in bands of 250 AED)



The mean and the median are marked on the histogram.

Q2 | What does the average hide?

a. Write one description of the histogram that the data support.

> Example: Costs are spread widely and skewed to the right: most businesses are between about 500 and 3,500 AED, with a long tail of high-cost businesses. The mean (1,862) is a little above the median (1,801).

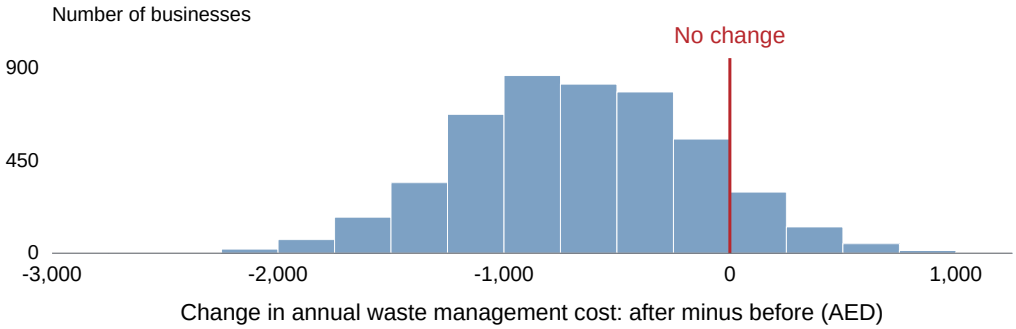
b. Write one thing that the mean alone cannot tell you.

> The mean cannot show how costs vary between businesses (spread, skew, unusual high-cost businesses), nor whether every business is close to the average.

Page 3 | Read two comparisons

H2-A3

Histogram of the change in annual waste management cost for the 4,794 participant businesses (AED)



Each business's change is its cost after minus its cost before. The histogram shows the changes that were recorded. It is not a confidence interval, and it does not show the effect of GreenWaste on each business.

Q3 | The same participant businesses, before and after

a. Use H2-A2 (the table of mean costs by group) on page 1. Calculate the change in mean cost for participant businesses, and write what it shows, with units and period.

> -669 AED: the participants' mean annual cost was 669 AED lower 12 months after than before, for the same 4,794 businesses.

b. Use the histogram. Did the cost fall for every business?

> No. Some bars are to the right of zero: 485 businesses' costs rose, 1 was unchanged and 4,308 fell. The mean fall does not describe every business.

c. Suppose waste-service prices changed during the year. This is a hypothetical example, not a fact about the case. What evidence would help separate a price change from the effect of GreenWaste?

> Examples: price or tariff changes for all businesses; changes in the amount of waste; other support programmes. Evidence: cost changes among comparable businesses that did not take part, or waste-service price records for the year.

Q4 | Comparison businesses and the difference after the programme

a. Calculate the change in mean cost for comparison businesses.

> +143 AED (2,401 - 2,258): the comparison businesses' mean cost rose.

b. Calculate the difference between the mean cost of participant businesses and of comparison businesses after the programme.

> -1,638 AED (763 - 2,401): participants' mean cost is 1,638 AED lower than comparison businesses' after the programme.

c. Explain why we cannot give GreenWaste the credit for the whole difference.

> The groups were not chosen by lottery and already differed before GreenWaste (1,432 vs 2,258), so part of the gap was there already. We do not know what participants would have paid without GreenWaste.

The two comparisons

Comparison	What it compares	What is missing
Before and after: -669 AED	The same participant businesses, cost after minus cost before	Their cost after a year without GreenWaste is not known.

Comparison	What it compares	What is missing
With and without: -1,638 AED	Participant businesses compared with comparison businesses, after a year	We do not know whether these two groups are a fair comparison.

Before GreenWaste, the mean cost was 1,432 AED for participant businesses and 2,258 AED for comparison businesses. Different recorded changes do not by themselves prove that GreenWaste caused a change, or show which way any bias goes.

Page 4 | Q5: Four decision cards

Use the cards and H2-A (the city data tables and charts). For each card: (a) choose Act, Ask first or Do not act; (b) write what the source supports; (c) write one question you would ask and why the answer matters; (d) say which action you mean.

Card 1

Decision (Act, Ask first or Do not act) and the action you mean: **Ask first (report the fall as a description; do not expand yet)**

What the source supports:

> The mean annual cost of the same participant businesses fell 669 AED (H2-A2). It is a correct description of before and after.

Your question and why the answer matters:

> Compared to what? What would the cost have been without GreenWaste (e.g. a fair comparison group)? It decides whether any of the fall can be credited to GreenWaste before expanding.

Card 2

Decision (Act, Ask first or Do not act) and the action you mean: **Do not act on the claim as worded (the 669 fall is real, but "effect" is not supported)**

What the source supports:

> The before-and-after interval (-684 to -654) correctly describes the mean change in those businesses. It does not show the programme effect, and 669 is below the 1,000 AED rule.

Your question and why the answer matters:

> Compared to what / how sure? There is no comparison group. Statistical precision is not attribution; ask for a fair comparison before calling it an effect.

Card 3

Decision (Act, Ask first or Do not act) and the action you mean: **Do not act (reject the national roll-out recommendation)**

What the source supports:

> $p < 0.001$ shows the mean change is unlikely to be exactly zero. It does not show size, cause, reach or value for money.

Your question and why the answer matters:

> Which businesses would it apply to, and what was the programme cost? A small p-value says nothing about whether other businesses nationwide would save at least 1,000 AED.

Card 4

Decision (Act, Ask first or Do not act) and the action you mean: **Ask first (use the two changes as a lead; check the comparison)**

What the source supports:

> Both changes are right: participants -669, comparison +143. The card gives no interval for the difference.

Your question and why the answer matters:

> How sure? How comparable are the two groups, and what is the interval for the difference between the changes? Without these we cannot judge the size of the difference caused by GreenWaste.

My final reading and request

Close the AI responses and turn over the decision cards, and keep H2-A2 (the table of mean costs by group) open.

a. Write one conclusion that the data support.

Example: Participant businesses' mean cost fell 669 AED while comparison businesses' mean cost rose 143 AED, but the groups differed before GreenWaste, so the data do not show how much was caused by the programme.

b. Write one specific piece of evidence you would request to help decide whether to expand GreenWaste.

Example: costs for both groups for earlier years, or a comparison of similar businesses just above and below the score cut-off, or the full programme cost. Accept any specific, relevant request.

c. Compare this with the reason you wrote on page 2. What changed?

Expect a move from "the fall shows the effect" to "the fall is a description; we need a fair comparison". Accept any honest reflection.

Page 5 | AI exercise

Part A: Mark the response

Naive prompt (a prompt with no context, as a busy person might type it): “Explain H2-A2 and advise whether to expand GreenWaste.” This response was written for the exercise. It is not a real AI answer. **Underline the passages that H2-A2 supports. Cross out the claims that H2-A2 does not support.**

> The chart shows a strong improvement: participant businesses' average annual waste cost fell from 1,432 to 763 AED, while costs for comparison businesses rose from 2,258 to 2,401 AED. The participant reduction of 669 AED is therefore unlikely to be just a city-wide fall in waste costs.
>
> After a year, participant businesses paid 1,638 AED less than comparison businesses. ~~This is the saving attributable to GreenWaste, because the comparison businesses show what would have happened without it.~~ ~~It exceeds the authority's 1,000 AED annual saving rule.~~
>
> The two groups began at different cost levels, so caution is appropriate. ~~However, the chart's diverging lines confirm that the programme worked across participant businesses.~~ ~~A wider rollout should deliver similar savings,~~ with monitoring to check delivery quality.

Part B: Run the improved prompt, then compare

Copy the prompt below and H2-A2 into Microsoft Copilot, and keep your AI good-practice card beside you. If you are not using Copilot, read the prepared response on page 6 after you finish Part A.

Improved prompt

Help a policy adviser read H2-A2 for a decision on expanding GreenWaste. The outcome is mean annual waste management cost per business in AED, measured before GreenWaste and 12 months after. The city has 4,794 participant businesses (score 58 or below) and 5,206 comparison businesses that did not receive GreenWaste (score above 58). They were selected by the score rule, not by a lottery. The groups started at different cost levels. Our rule is a reduction of at least 1,000 AED in annual waste management cost per business per year, caused by the programme.

Ask any essential clarifying questions first. Then give three short points: the before-and-after change and the difference between the two groups after the programme; what these comparisons cannot show; and one specific piece of evidence to request before making a recommendation. Cite H2-A2. Do not assume that the comparison businesses show what would have happened to participant businesses without GreenWaste. Do not assume that a mean describes every business. Do not assume that these results apply nationwide.

a. What improved compared with Part A?

> Look for: it separates the before-and-after change from the gap after the programme; it asks about comparability; it no longer calls 1,638 "the saving"; and it names a specific piece of evidence.

b. Which claims are still unsupported?

> Likely still unsupported: any statement that comparison businesses show what would have happened; national transfer; that every business improved; that the 1,000 AED rule is met.

c. Which table or figure should you check each claim against?

> Check numbers (1,432, 763, 2,258, 2,401, -669, +143, -1,638) against H2-A2; check claims about causes and the rule against the "Neither comparison alone..." note and the rule text.

Page 6 | Prepared response

Use this response if you are not using Copilot. Read it after you have marked the response on page 5. It is an example of a better answer. An AI tool may answer differently.

> Before I advise on expansion, I would ask how the efficiency scores were determined, whether other changes affected the two groups differently, and what evidence supports using the comparison businesses to estimate participant businesses' costs without GreenWaste.
>
> H2-A2 records a 669 AED mean fall among participant businesses and a 143 AED mean rise among comparison businesses. After the programme, the participant businesses' mean cost is 1,638 AED lower than the comparison businesses' mean cost. But the participant businesses already had lower mean costs before the programme (1,432 AED, compared with 2,258 AED). Neither the before-and-after change nor the difference between the groups after the programme is, on its own, a credible estimate of a saving caused by GreenWaste.
>
> The group means do not show that every business improved or that national expansion would produce the same result. Before I judge whether GreenWaste meets our rule of at least a 1,000 AED reduction in annual waste management cost per business per year, I would request evidence for a fair comparison. The chart is useful descriptive evidence, but it is not enough by itself for a decision on expansion.

Check it against H2-A2 (the table of mean costs by group), as you would check a live AI answer.

a. Does it separate what was observed, what a fair comparison would need, and what decision to make?

> Yes: observed (669 fall, 143 rise, 1,638 gap), what a fair comparison needs (different starting levels; comparability), and the decision (not enough yet).

b. Which clarifying question in the response is most helpful?

> Example: "what evidence supports using the comparison businesses to estimate participant businesses' costs without GreenWaste". Accept any with a reason.

c. What evidence is still needed before advising on expansion?

> Costs of both groups before GreenWaste over several periods, how scores were set, other changes that hit groups differently, and programme costs.

Page 7 | Check the data before you calculate

These questions go with the R slides on the file, the filter and the cleaning examples.

Q6 | What is in the file?

a. How many rows and columns does the GreenWaste file have, and what does each row represent?

> 10,400 rows and 12 columns. Each row is one business.

b. How many of the rows are city businesses and how many are pilot businesses?

> 10,000 city businesses and 400 pilot businesses.

Q7 | Which businesses are in the calculation?

a. Why do we keep only the city rows before we calculate the city averages?

> The pilot is a separate study (lottery, different population). Mixing it in changes who is in each group and so the means.

b. A colleague included the pilot businesses by mistake. How could you tell from the number of participant businesses?

> The group sizes would be 4,994 and 5,406 instead of 4,794 and 5,206 (the 200 pilot treatment and 200 waiting businesses added), even though the results look similar.

Q8 | Invented cleaning examples

a. Choose one of the four problems on the slide. Write what went wrong and how it changes the answer.

> Any one: duplicate business counted twice (mean change -440.0 becomes -450.0); missing-value code -99 treated as a cost (mean 1,550.0 falls to 1,220.2); 1.7 typed instead of 1,700 (mean 1,580.0 falls to 1,240.3); four group labels instead of two, so each group mean describes only some businesses.

b. Why is a wrong structure in the data more dangerous than a calculation that stops with an error message?

> None of the four gives an error message: the calculation runs and returns a plausible-looking number, so the mistake can be reported without anyone noticing.